

Machine Learning (ML) in Hindi

Week-4 Assignment

Total points: 15

Topics covered:

- Logistic Regression
- Classification evaluation metrics

Instructions

- Each question is a 1-point multiple choice question (MCQ) with only one correct answer unless specified otherwise.
- Multiple select questions (MSQ) are 2-point questions with one or more than one answers correct
- Numerical-answer type questions carry 2-points and have an exact number. Try to be as precise as possible.
- For programming questions, use the default parameter settings unless specified otherwise.

1. [MSQ] What are the different types of encoding techniques to handle categorical data?
 - a) One-Hot Encoding
 - b) Label Encoding
 - c) Ordinal Encoding
 - d) Numerical Value Imputation

Answer: a), b), c)

2. [MSQ] What are the different types of solver methods available in the class LogisticRegression in the Scikit-Learn Library?
 - a) lbfgs
 - b) liblinear
 - c) newton-cg
 - d) sag

Answer: a), b), c), d)

3. [MCQ] Consider the task of classifying e-mails as either spam or non-spam. Which of the following models is most suitable for this spam detection problem?
 - a) Binary Logistic Regression Model
 - b) Multi-Class Linear Regression Model

- c) Linear Regression Model
- d) k-NN Regression Model

Answer: a) Binary Logistic Regression Model

4. [MCQ] What is the range of the tanh function that is used for classification tasks?
- a) (-1,1]
 - b) (0,1)
 - c) [0,1]
 - d) [-1,1]

Answer d) [-1,1]

5. [MCQ] What loss function do we use for multi-class classification in training a Logistic Regression model?
- a) Smooth L1 Loss
 - b) Cross-Entropy Loss
 - c) Binary Cross-Entropy Loss
 - d) Root Mean Square Error

Answer: b)

6. [MCQ] Which of the following activation function can be used in the logistic regression for two-class classification problem?
- a) Sigmoid
 - b) ReLU
 - c) Cosine
 - d) Sine

Answer: a) Sigmoid

7. [MSQ] Which of the following is/are possible values that the “penalty” can assume in the logistic regression function of the sklearn library?
- a) “L1”
 - b) “L2”
 - c) “L0”
 - d) “Elasticnet”

Answer: a), b), d)

8. [MCQ] Which of the following is the correct formula for the binary cross-entropy loss function?

a) $L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$

b) $L = \frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 + y_i) \log(1 - p_i)]$

c) $L = -\sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(p_i)]$

d) $L = -\frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$

Answer: a) $L = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$

9. [MCQ] In the exam passing example, what is the probability of passing if a student studies for 3.5 hours, when $\hat{y} = \frac{1}{1+e^{-z}}$, where $z = -1 + 0.5 * \text{study hours}$?

a) 0.90

b) 0.78

c) 0.59

d) 0.68

Answer – d) 0.68

10. [MCQ] From the given below data for actual labels and predicted labels from a machine learning model for a two-class classification problem, compute accuracy for the machine learning model. Consider Class 1 as positive label and Class 0 as negative label.

Actual Labels	Predicted Labels
Class 0	Class 0
Class 1	Class 0
Class 0	Class 1
Class 1	Class 0
Class 1	Class 1
Class 0	Class 1
Class 1	Class 1
Class 0	Class 0
Class 0	Class 0

a) 94%

b) 38%

c) 55%

d) 62%

Answer – c) 55%

11. [MSQ] Using the table from the above question, calculate specificity and sensitivity of the machine learning model

- a) Sensitivity – 40%
- b) Specificity – 85%
- c) Sensitivity – 80%
- d) Sensitivity – 90%
- e) Specificity – 60%
- f) Specificity – 100%
- g) Sensitivity – 50%
- h) Specificity – 55%

Solution: (e), (g)